Comparing different types of GPU memories

Comparison Charts

Fig. 1: Best speedup by matrix width

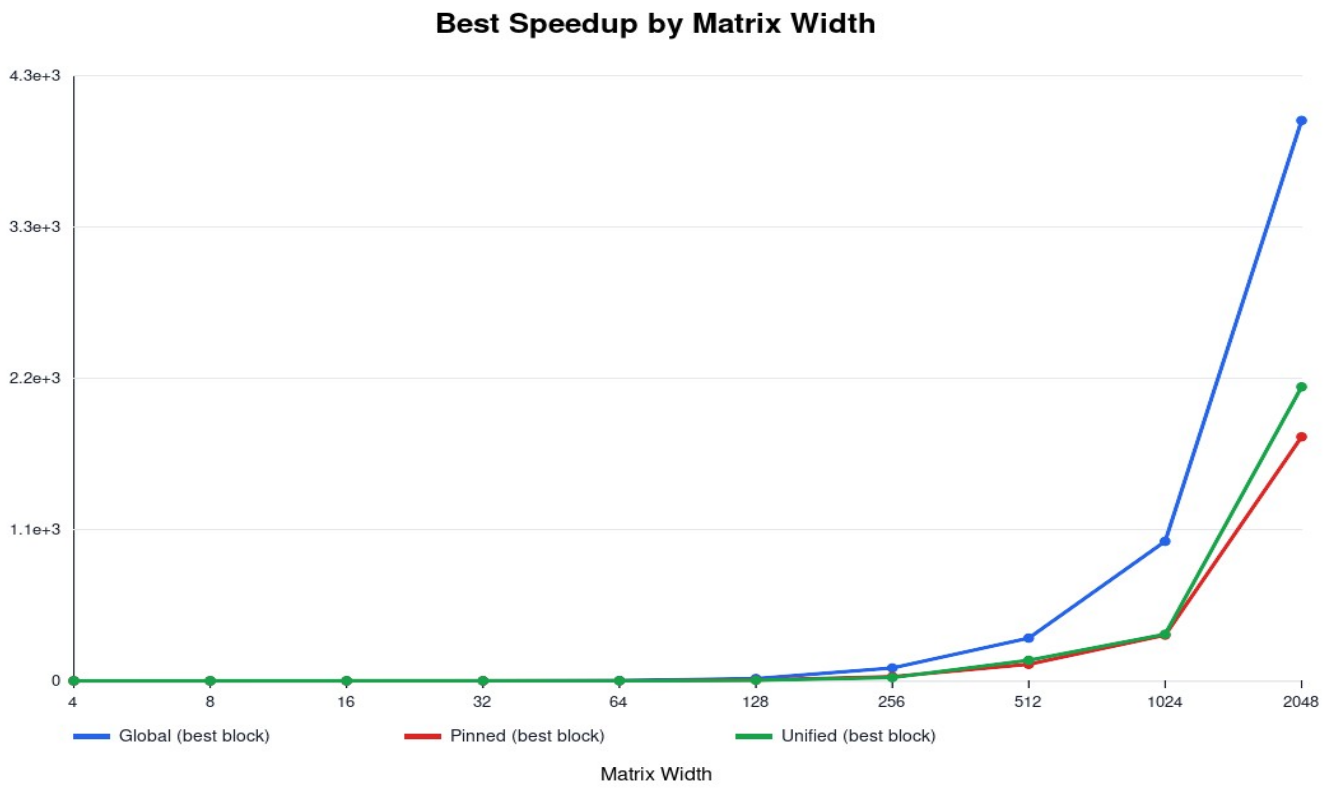


Fig. 2: Best GPU total time by matrix width

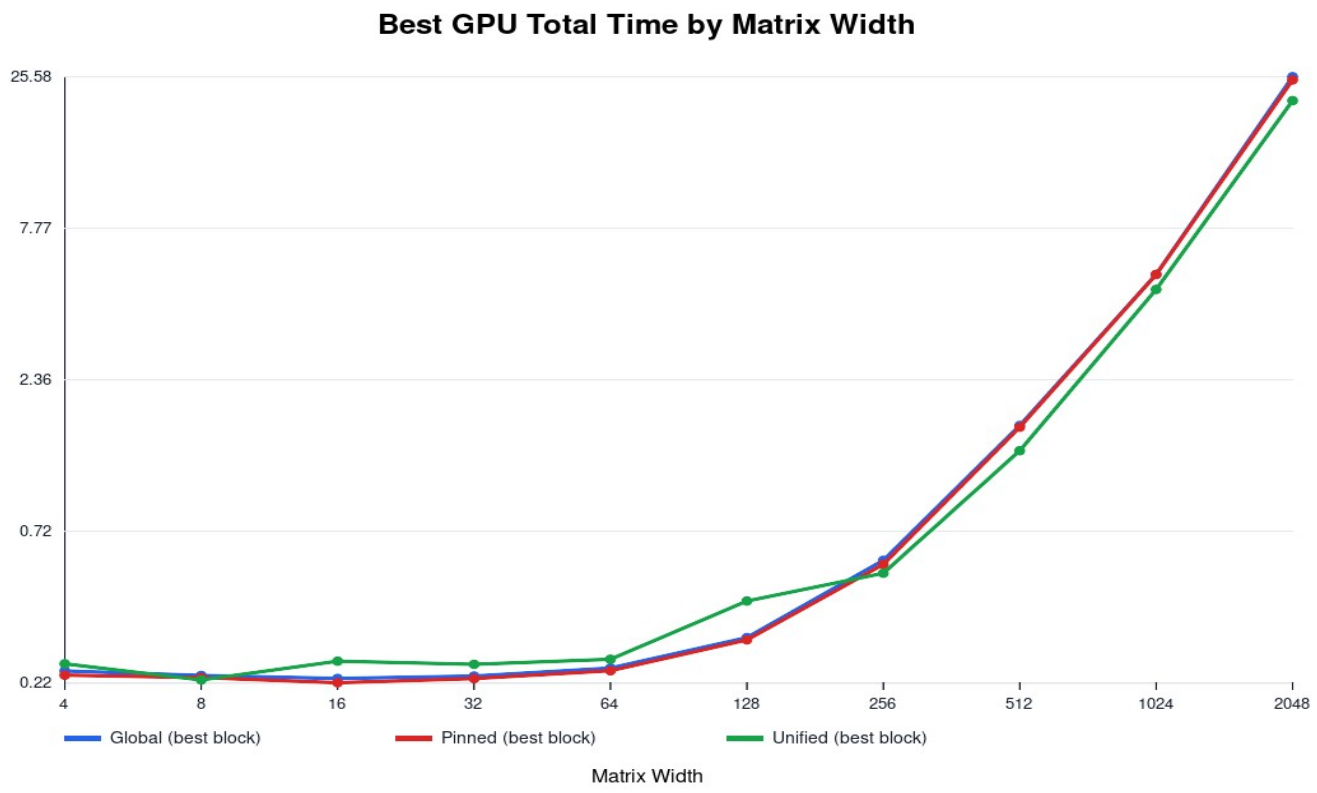


Fig. 3: GPU total time at block size 16

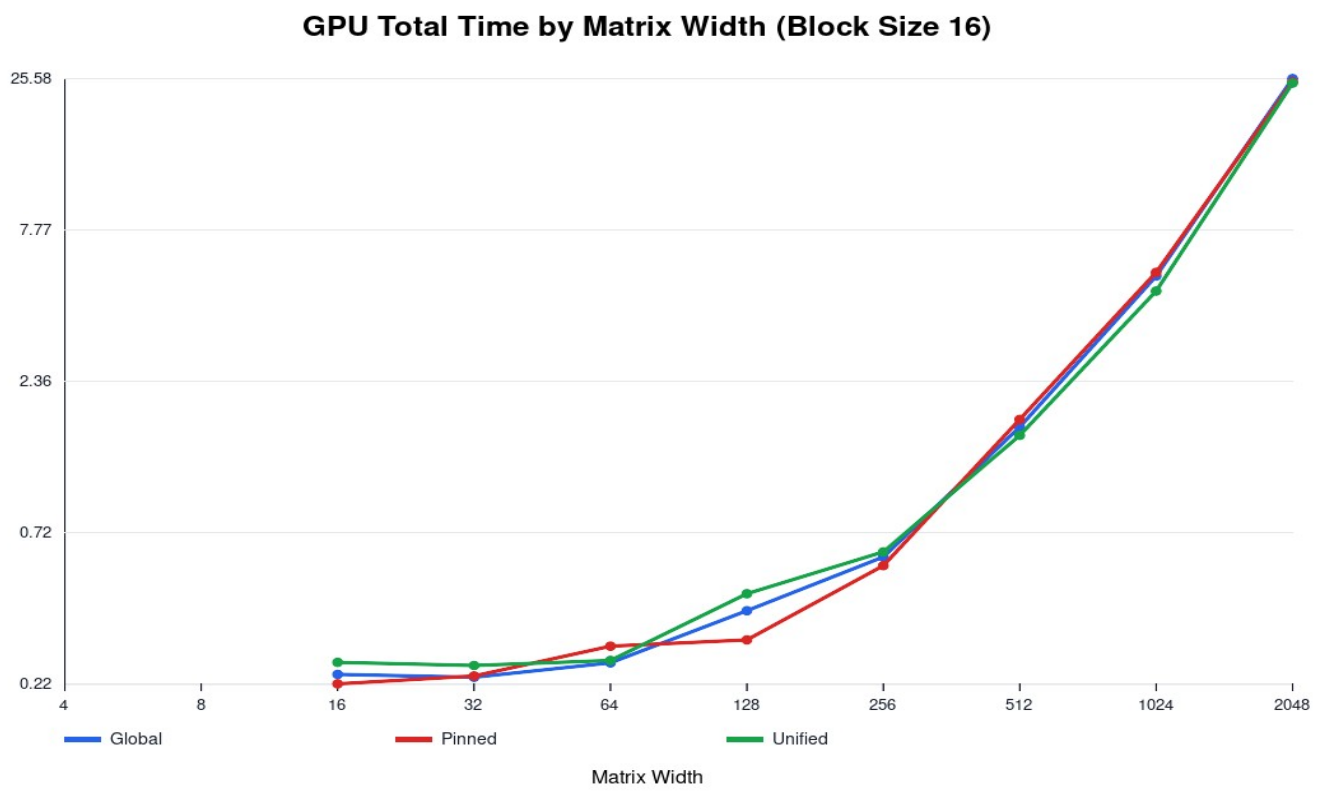


Fig. 4: Transfer time at block size 16

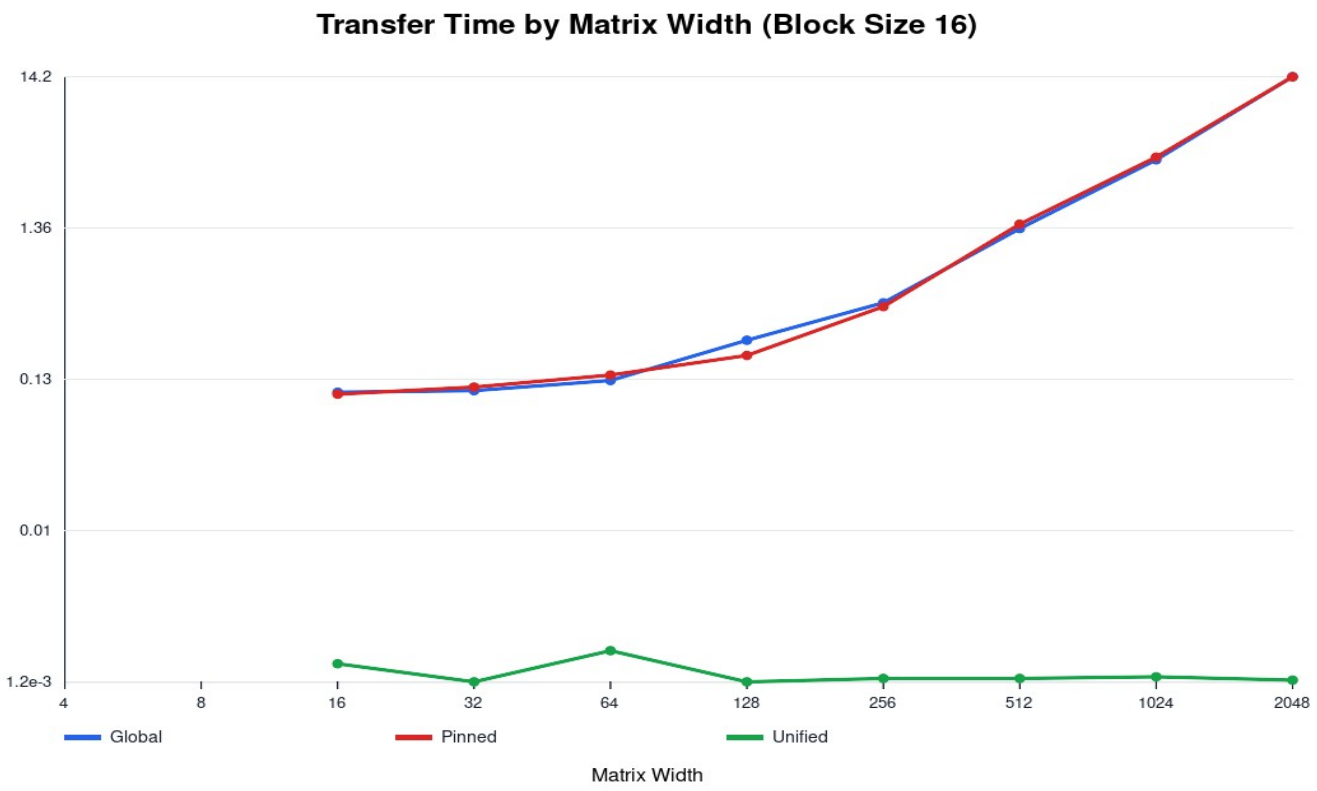


Fig. 5: Block size with best speedup

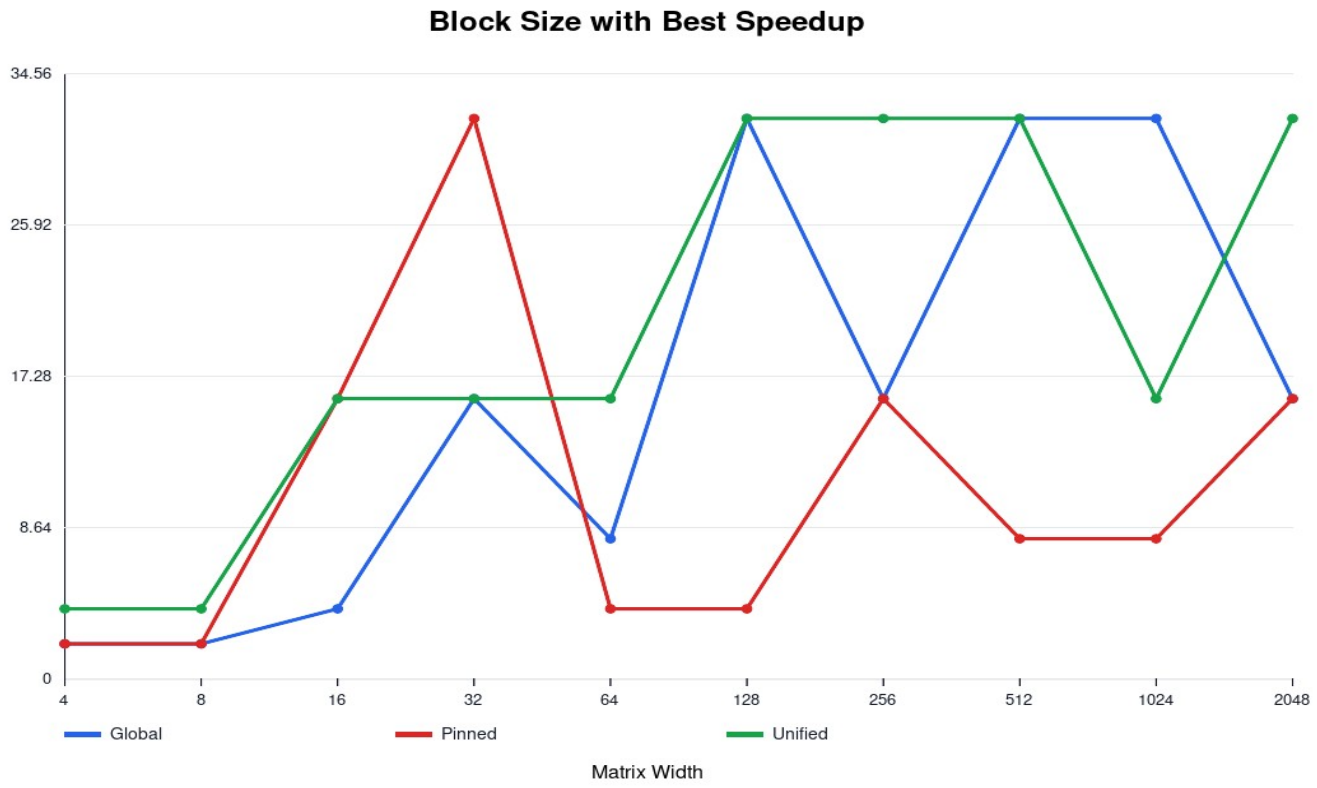
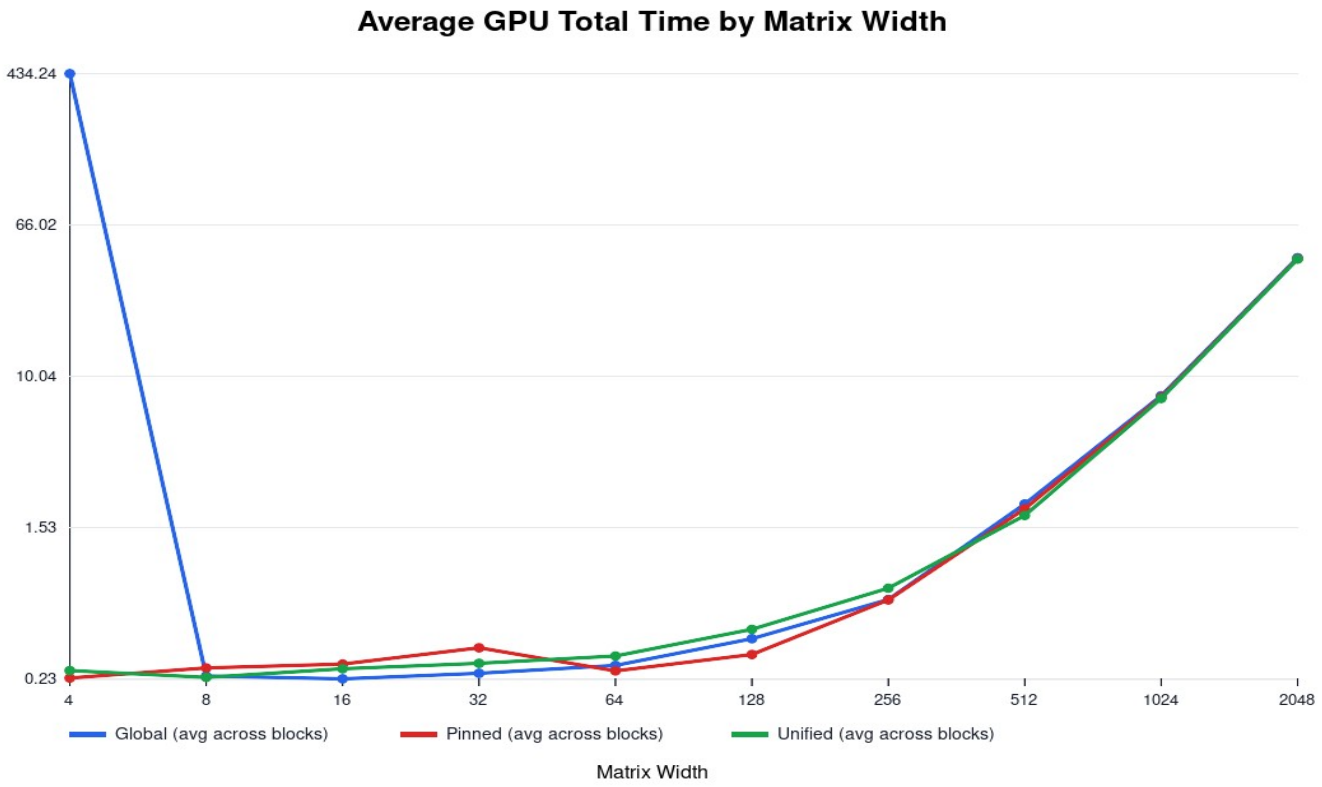


Fig. 6: Average GPU total time by matrix width (avg over block sizes)



Findings

For small widths from around 4 to 32, speedup is around zero or below most of the times. This is because the overhead of launching combined with memory overhead is bigger than the actual computational work.

At block sizes above or equal to 512, all three versions accelerate by quite lot. Best speedups I measured:

- `Global`: `4026.726x` at width `2048`, block `16`
- `Pinned`: `1753.446x` at width `2048`, block `16`
- `Unified`: `2112.387x` at width `2048`, block `32`

3) Unified reports dramatically lower transfer time (probably bugged)

Unified shows transfer times around `~0.0012 ms` across sizes, while global and pinned rise with matrix width (up to about `14 ms` at width `2048`).

That lines up with what I'd expect from unified memory since copies are mostly hidden compared to regular host-device copies since it works in the background and tries to be as non-intrusive as possible. I'd need to look into the actual architecture and algorithms, but I'd expect it to not be a stop-the-world type scenario and more of a background/parallel process that fetches memory.

Looking at best-case GPU total time by width:

- Width `512`: Unified best `1.349 ms` (block `32`)
- Width `1024`: Unified best `4.799 ms` (block `16`)
- Width `2048`: Unified best `21.199 ms` (block `32`)

Pinned and global are close in some places, but unified typically wins in these larger cases. Again, it's fairly even in the smaller cases because the computation time essentially doesn't matter because we end up benching the transfer time instead

As matrix width increases, the best speedup usually comes from block size `16` or `32`. Smaller blocks (`2`, `4`) generally leave performance on the table. It's interesting how sometimes the best size is 16 since i'd expect 32 to be better since it fills an entire warp.

tl;dr, from both the charts and data tables, i learned that - For large `n`, matrix multiply has enough work to fully use GPU parallelism - For small `n`, fixed overhead dominates - Compute work grows as `O(n^3)`, while transfer grows at a rate closer to `O(n^2)`, so bigger matrices end up favoring the GPU more and more - Block sizes around `16/32` tend to give better utilization/occupancy

Note: speedup uses each run's own CPU baseline. Since CPU times differ across the three result files, direct speedup-vs-speedup comparisons across modes are not perfectly apples-to-apples.

So for strict cross-mode comparison, I'd trust `GPU Total Time` and `Transfer Time` more than raw speedup alone.

In conclusion, if I care most about end-to-end GPU time in this dataset, `Unified` is usually the best choice. At the same time, `Pinned` and `Global` are still strong, just generally a bit slower in best-case large-width runs and for tiny matrices, CPU is always the winner.

Tables

Global Memory:

Matrix Width	Block Size	CPU Time (ms)	GPU Total Time (ms)	Total Data Transfer Time (ms)	GPU Processing Time (ms)	Speedup
4	2	0.000000	0.756398	0.666598	0.089800	0.000000
4	4	0.000000	0.739494	0.662689	0.076805	0.000000
8	2	0.000000	0.664874	0.586016	0.078858	0.000000
8	4	0.000000	0.667830	0.597748	0.070082	0.000000
8	8	0.000000	0.832761	0.761036	0.071725	0.000000
16	2	0.001000	0.690573	0.624598	0.065975	0.001448
16	4	0.001000	0.752819	0.675093	0.077726	0.001328
16	8	0.001000	0.744264	0.667359	0.076905	0.001344
16	16	0.001000	0.737481	0.664964	0.072517	0.001356
32	2	0.008000	0.766417	0.684843	0.081574	0.010438
32	4	0.010000	0.778948	0.701302	0.077646	0.012838
32	8	0.007000	0.745958	0.666287	0.079671	0.009384
32	16	0.008000	0.667159	0.588931	0.078228	0.011991
32	32	0.008000	0.735968	0.654494	0.081474	0.010870
64	2	0.071000	0.852197	0.767427	0.084770	0.083314
64	4	0.063000	0.852156	0.765683	0.086473	0.073930
64	8	0.063000	0.945543	0.854852	0.090691	0.066628
64	16	0.063000	0.866315	0.762248	0.104067	0.072722
64	32	0.063000	0.872094	0.779840	0.092254	0.072240

Matrix Width	Block Size	CPU Time (ms)	GPU Total Time (ms)	Total Data Transfer Time (ms)	GPU Processing Time (ms)	Speedup
128	2	0.986000	1.366146	1.216033	0.150113	0.721738
128	4	0.960000	1.417413	1.302837	0.114576	0.677290
128	8	0.958000	1.856060	1.373821	0.482239	0.516147
128	16	0.970000	1.362289	1.262461	0.099828	0.712037
128	32	0.957000	1.343715	1.244287	0.099428	0.712205
256	2	10.447000	3.731197	3.107340	0.623857	2.799906
256	4	10.397000	3.218479	2.978887	0.239592	3.230408
256	8	10.507000	3.300614	3.131385	0.169229	3.183347
256	16	10.522000	3.255699	3.086310	0.169389	3.231871
256	32	10.547000	4.147119	3.969855	0.177264	2.543211
512	2	220.403000	5.928600	1.424075	4.504525	37.176230
512	4	220.725998	2.644085	1.426098	1.217987	83.479161
512	8	221.992996	2.098829	1.438974	0.659855	105.769930
512	16	221.548996	2.110650	1.461215	0.649435	104.967188
512	32	221.973007	2.074020	1.432601	0.641419	107.025490
1024	2	4408.433105	37.693359	2.318352	35.375008	116.955167
1024	4	3840.277100	11.423080	2.254531	9.168550	336.185784

Matrix Width	Block Size	CPU Time (ms)	GPU Total Time (ms)	Total Data Transfer Time (ms)	GPU Processing Time (ms)	Speedup
1024	8	4119.008789	7.051518	2.255413	4.796105	584.130791
1024	16	4476.875000	7.079230	2.241126	4.838104	632.395755
1024	32	4448.898926	7.221358	2.328852	4.892506	616.075110
2048	2	40972.187500	390.832550	8.079847	382.752716	104.833099
2048	4	41602.996094	106.239380	9.026371	97.213005	391.596752
2048	8	41371.207031	57.394989	8.388709	49.006283	720.815663
2048	16	41386.089844	58.807564	9.327328	49.480236	703.754535
2048	32	41131.183594	57.685371	8.069076	49.616295	713.026247

Pinned Memory

Matrix Width	Block Size	CPU Time (ms)	GPU Total Time (ms)	Total Data Transfer Time (ms)	GPU Processing Time (ms)	Speedup
4	2	0.000000	0.237952	0.121760	0.116192	0.000000
4	4	0.000000	0.230848	0.115680	0.115168	0.000000
8	2	0.000000	0.226496	0.113344	0.113152	0.000000
8	4	0.000000	0.230208	0.114816	0.115392	0.000000
8	8	0.000000	0.340608	0.154464	0.186144	0.000000
16	2	0.001000	0.286560	0.153152	0.133408	0.003490

Matrix Width	Block Size	CPU Time (ms)	GPU Total Time (ms)	Total Data Transfer Time (ms)	GPU Processing Time (ms)	Speedup
16	4	0.001000	0.288224	0.145184	0.143040	0.003470
16	8	0.001000	0.323040	0.202080	0.120960	0.003096
16	16	0.001000	0.217536	0.102784	0.114752	0.004597
32	2	0.008000	0.688352	0.542144	0.146208	0.011622
32	4	0.008000	0.274304	0.148576	0.125728	0.029165
32	8	0.008000	0.288832	0.151872	0.136960	0.027698
32	16	0.007000	0.231424	0.114816	0.116608	0.030248
32	32	0.008000	0.224736	0.108192	0.116544	0.035597
64	2	0.063000	0.266432	0.141120	0.125312	0.236458
64	4	0.063000	0.239072	0.122432	0.116640	0.263519
64	8	0.063000	0.240544	0.123872	0.116672	0.261906
64	16	0.063000	0.292704	0.138400	0.154304	0.215235
64	32	0.064000	0.241920	0.120128	0.121792	0.264550
128	2	0.987000	0.326592	0.189216	0.137376	3.022119
128	4	1.648000	0.304640	0.180320	0.124320	5.409664
128	8	0.960000	0.317728	0.189568	0.128160	3.021452
128	16	0.986000	0.307520	0.187488	0.120032	3.206296
128	32	1.040000	0.314304	0.186720	0.127584	3.308898
256	2	10.832001	0.728608	0.439040	0.289568	14.866706
256	4	11.279000	0.579264	0.410752	0.168512	19.471260

Matrix Width	Block Size	CPU Time (ms)	GPU Total Time (ms)	Total Data Transfer Time (ms)	GPU Processing Time (ms)	Speedup
256	8	10.857000	0.578976	0.424608	0.154368	18.752073
256	16	10.853001	0.551936	0.400192	0.151744	19.663514
256	32	12.146001	0.661312	0.485920	0.175392	18.366521
512	2	224.709015	2.780992	1.334336	1.446656	80.801748
512	4	225.917007	1.826144	1.290816	0.535328	123.712592
512	8	224.841003	1.623712	1.291968	0.331744	138.473450
512	16	223.557007	1.746272	1.438624	0.307648	128.019579
512	32	225.927017	1.668640	1.349408	0.319232	135.395901
1024	2	4500.420410	14.577663	3.883648	10.694016	308.720294
1024	4	4290.791992	7.409568	4.005920	3.403648	579.088011
1024	8	4486.427246	5.720799	3.949664	1.771136	784.230882
1024	16	4512.746094	5.560864	4.070016	1.490848	811.518874
1024	32	4530.449219	5.398944	3.929024	1.469920	839.136175
2048	2	41839.339844	98.906113	13.965120	84.940994	423.020768
2048	4	41929.761719	40.572094	14.414880	26.157215	1033.463092
2048	8	41867.664062	27.631361	14.476992	13.154368	1515.222651
2048	16	42329.507812	24.944321	14.160448	10.783872	1696.959713
2048	32	41171.89062	25.003328	14.263648	10.739680	1646.65642

Matrix Width	Block Size	CPU Time (ms)	GPU Total Time (ms)	Total Data Transfer Time (ms)	GPU Processing Time (ms)	Speedup
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Unified Memory

Matrix Width	Block Size	CPU Time (ms)	GPU Total Time (ms)	Total Data Transfer Time (ms)	GPU Processing Time (ms)	Speedup
4	2	0.000000	0.262016	0.001760	0.260256	0.000000
4	4	0.000000	0.252416	0.001184	0.251232	0.000000
8	2	0.000000	0.235776	0.001184	0.234592	0.000000
8	4	0.000000	0.222304	0.001152	0.221152	0.000000
8	8	0.000000	0.250848	0.001184	0.249664	0.000000
16	2	0.001000	0.270816	0.001152	0.269664	0.003693
16	4	0.001000	0.263904	0.001184	0.262720	0.003789
16	8	0.001000	0.259840	0.001184	0.258656	0.003849
16	16	0.001000	0.257664	0.001568	0.256096	0.003881
32	2	0.017000	0.304192	0.001408	0.302784	0.055886
32	4	0.019000	0.292960	0.001472	0.291488	0.064855
32	8	0.018000	0.300000	0.001184	0.298816	0.060000
32	16	0.018000	0.251392	0.001184	0.250208	0.071601
32	32	0.021000	0.259008	0.001152	0.257856	0.081079
64	2	0.073000	0.325824	0.001184	0.324640	0.224047
64	4	0.072000	0.349024	0.001184	0.347840	0.206290
64	8	0.072000	0.328896	0.001184	0.327712	0.218914

Matrix Width	Block Size	CPU Time (ms)	GPU Total Time (ms)	Total Data Transfer Time (ms)	GPU Processing Time (ms)	Speedup
64	16	0.074000	0.261568	0.001920	0.259648	0.282909
64	32	0.071000	0.276128	0.001184	0.274944	0.257127
128	2	0.980000	0.434752	0.001184	0.433568	2.254159
128	4	0.977000	0.428672	0.001312	0.427360	2.279132
128	8	0.975000	0.429504	0.001184	0.428320	2.270060
128	16	1.019000	0.442464	0.001184	0.441280	2.303012
128	32	0.971000	0.413600	0.001216	0.412384	2.347679
256	2	10.512000	0.983872	0.001408	0.982464	10.684317
256	4	10.559001	0.841568	0.001248	0.840320	12.546819
256	8	10.898001	0.630560	0.001216	0.629344	17.283052
256	16	10.408001	0.615232	0.001248	0.613984	16.917197
256	32	10.196000	0.514752	0.001216	0.513536	19.807597
512	2	208.789017	2.564928	0.001248	2.563680	81.401512
512	4	210.497009	1.787904	0.001216	1.786688	117.733955
512	8	208.750015	1.625728	0.001344	1.624384	128.404023
512	16	210.820007	1.540448	0.001248	1.539200	136.856296
512	32	214.034012	1.349344	0.001280	1.348064	158.620791
1024	2	5026.595215	14.988064	0.001216	14.986848	335.373215
1024	4	4558.365234	7.745536	0.001344	7.744192	588.515144
1024	8	5024.473145	5.707712	0.001216	5.706496	880.295492

Matrix Width	Block Size	CPU Time (ms)	GPU Total Time (ms)	Total Data Transfer Time (ms)	GPU Processing Time (ms)	Speedup
1024	16	4550.825195	4.799489	0.001280	4.798208	948.189525
1024	32	5040.452148	4.803456	0.001248	4.802208	1049.338674
2048	2	51101.250000	101.583298	0.001216	101.582077	503.047755
2048	4	51399.156250	42.019615	0.001216	42.018398	1223.218163
2048	8	51066.832031	26.174465	0.001504	26.172960	1951.017224
2048	16	51162.218750	24.703617	0.001216	24.702400	2071.041611
2048	32	50925.363281	21.199135	0.001248	21.197887	2402.237793